Definition

A *presentation* $\langle \mathbf{G}, s, t, \mathbf{R} \rangle$ of a prop is a set $\mathbf{G}$, source and target functions $s, t : \mathbf{G} \to \mathbb{N}$, and a relation $\mathbf{R} \subseteq E(\mathbf{G}) \times E(\mathbf{G})$ between expressions such that for any $\langle e_1, e_2 \rangle \in \mathbf{R}$, the expressions e_1 and e_2 have the same arity.

The prop $\mathbf{C}$ *presented* by $\langle \mathbf{G}, s, t, \mathbf{R} \rangle$ is the prop whose morphisms are elements of $E(\mathbf{G})$ quotiented by the equations (or "relations") $e_1 = e_2$ for $\langle e_1, e_2 \rangle \in \mathbf{R}$, and by the axioms for symmetric strict monoidal categories.